Relative path: contracts/schemas.py
Filename: schemas.py Extension: .py

```
"""
OASIS v2 — shared data contracts.

Every layer imports its input/output shapes from here. Nobody is allowed to
silently drift a field name mid-build — if a layer needs a new field, it gets
added here first, and every downstream layer that consumes it gets updated in
the same commit.

NOTE on `label` / `anomaly_type`: these fields exist ONLY on the ground-truth
side channel (see data_gen/generate_logs.py -> ground_truth.json). They are
never present on the model-facing Session objects the detection/classification
layers actually see. AccessEvent/Session below intentionally have no label
field for that reason.
"""
from __future__ import annotations

from enum import Enum
from typing import List, Optional

from pydantic import BaseModel, Field


class EntityType(str, Enum):
    user = "user"
    service_account = "service_account"
    edge_device = "edge_device"


class AttackType(str, Enum):
    """The 7 required attack types + 1 edge case (insider_drift), per the
    brief's suggested taxonomy and ARCHITECTURE.md Section 1/3."""
    brute_force = "brute_force"
    impossible_travel = "impossible_travel"
    credential_stuffing = "credential_stuffing"
    lateral_movement = "lateral_movement"
    low_and_slow = "low_and_slow"
    privilege_escalation = "privilege_escalation"
    device_spoofing = "device_spoofing"
    # Edge case: NOT an attack. A legitimate behavioral shift (new work
    # pattern / new device) used to test that concept drift does not result
    # in a permanently-flagged entity. Ground truth label = "normal".
    insider_drift = "insider_drift"


class AccessEvent(BaseModel):
    """One raw access/connection event. Matches the brief's suggested schema."""
    event_id: str
    entity_id: str
    entity_type: EntityType
    timestamp: str  # ISO8601
    source_ip: str
    geo_location: str
    resource_accessed: str
    auth_method: str
    action: str
    auth_success: bool
    device_fingerprint: str
    session_id: str


class Session(BaseModel):
    """The unit of analysis (ARCHITECTURE.md Section 1). One login-to-logout
    span per entity. This is what both the profiling and detection models
    actually score. NO label field on purpose — ground truth lives only in
    ground_truth.json."""
    session_id: str
    entity_id: str
    entity_type: EntityType
    start_time: str
    end_time: str
    session_duration_sec: float
    command_sequence: List[str]         # ordered actions taken
    resources_touched: List[str]
```

```
    auth_failures: int
    source_ips: List[str]
    geo_locations: List[str]
    device_fingerprint: str
    is_new_device: bool


class AnomalyEvent(BaseModel):
    """Output of Layer 0 (profiling, rarity_score) + Layer 1 (detection,
    anomaly_score). Combined here because the detection layer's fallback path
    (cold-start rule engine) IS the profiling layer's cold-start signal."""
    session_id: str
    entity_id: str
    anomaly_score: float               # 0-1, higher = more anomalous
    rarity_score: float                # 0-1, from baseline profiler
    is_cold_start: bool
    used_fallback_rule: bool           # True if scored by rule engine, not GRU
    contributing_features: List[str]   # ordered, most-important first


class ClassifiedAnomaly(BaseModel):
    """Output of Layer 2. Only ever computed for sessions Layer 1 flagged."""
    session_id: str
    anomaly_type: AttackType
    type_confidence: float
    class_probabilities: dict


class RiskAssessment(BaseModel):
    """Output of Layer 3. Transparent arithmetic only — no model here."""
    session_id: str
    entity_id: str
    anomaly_score: float
    anomaly_type: AttackType
    type_confidence: float
    mitre_technique_id: Optional[str]
    mitre_technique_name: Optional[str]
    asset_criticality: int
    impact_score: float                # 0-1
    contributing_features: List[str]
    recommended_action: str            # acknowledge / isolate / escalate


class IncidentNarrative(BaseModel):
    """Output of Layer 4 (not built yet — placeholder contract so Layer 3's
    output shape is already the exact thing Layer 4 will consume)."""
    session_id: str
    summary: str
    why_flagged: str
    recommended_action: str
    confidence: float
```